

BRIGHTLEARN DATA ANALYTICS

SQL FUNDAMENTALS ASSIGNMENT 1

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Course: Brightlearn Basic SQL

SQL fundamentals: SELECT & filtering

Section 1 Selecting and Sorting

Q1. All columns from the employees table

SELECT *

FROM employees;

employees

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Carthy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q2. All unique departments

SELECT DISTINCT department

FROM employees;

department
IT
HR
Finance
Marketing

Q3. First and last names ordered by salary, descending

SELECT first_name, last_name

FROM employees

ORDER BY salary DESC;

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Q4. The top 3 highest-paid employees

```
SELECT id, first-name, last-name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

id	first-name	last-name	Salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

SECTION 2. Filtering with WHERE

Q5. Employees in the IT department

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'IT';
```

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Q6. Employees in the Finance with Salary > 60000

```
SELECT id, first-name, last-name, department, salary
FROM employees
WHERE department = 'Finance'
AND salary > 60000
```

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65000

Q7. Employees in HR or Marketing

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'HR'
OR department = 'Marketing';
```

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q8. employees not in IT

```
SELECT id, first-name, last-name, department
FROM employees
WHERE NOT department = 'IT';
```

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Q9. employees in IT, HR, or Finance using IN

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Q10 employees in IT with salary > 65000 and city Johannesburg.

```
SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';
```

id	first_name	last_name	department	Salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg